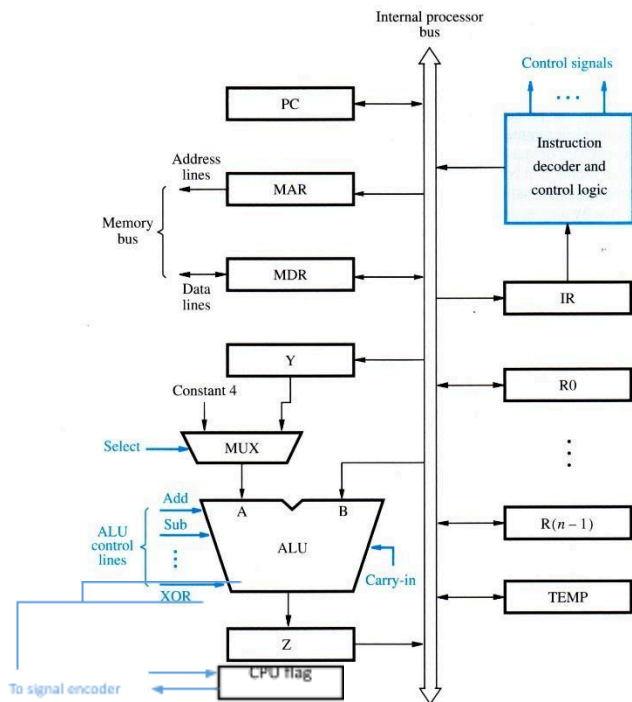


CPE223 Computer Architecture

PS7 Processor Design

Introduction

Instruction set can be divided into four main categories including arithmetic/logic operation, transfer data, fetch/store data, and branch with several addressing modes to access the memory. In this lab, we are going to figure out the stages and data paths of each instructions and find out how to design a hardware control for the processor to perform some selected instructions.



Instruction

ADD	Rn,off(Rn)
ADD	Rn,#value
ADD	Rn,Rn
MOV	Rn,(Rn)
MOV	(index),Rn
BZ	(Rn)
BZ	+#relative
B	#Absolute

Figure 1 Single bus processor

For our lab, we will use single bus system as illustrated in figure 1. Then we will use only a few of them to grasp the idea on how to design the stages and data paths. The instruction set is listed here consists of add, move, and branch operations.

Task 1

design stages and data paths of each instruction to get the idea on the effect of addressing mode and complex instruction on the stages and data paths. There are eight instructions to divide among your group members to come up with stages and data paths.

Ex. Add (R3), R1

Step	Action
1	PC_{out} , MAR_{in} , Read, Select4, Add, Z_{in}
2	Z_{out} , PC_{in} , WMFC
3	MDR_{out} , IR_{in}
4	$R3_{out}$, MAR_{in} Read
5	$R1_{out}$, Y_{in} , WMF
6	MDR_{out} , SelectY, Add, Z_{in}
7	Z_{out} , $R1_{in}$, End

Task 2

design a hardwire circuit to generate control signal for Y_{in} , Z_{in} , R_{in} , and MAR_{in} from the selected instruction from the previous task. The instructions that you have to consider are three ADDs, and BZ (R_n).

Report submission

create a report consists of your design of the first task and the design of the second tasks, please show the design result and discuss the number of stages and how complex your data paths are. Submit it in PDF format via LEB2.

Deadline: 10th March 2026 23:59.

This is the problem session for the group of 4-6 students.